Exercise-3

Operators and Expressions, Variables and Type conversions

a. Evaluate the following expressions

i. a/b*c-b+a*d/3

```
#include<stdio.h>
int main()
{
   int a, b, c, d, result;
   scanf("%d %d %d %d", &a, &b, &c, &d);
   result = a / b * c - b + a * d / 3;
   printf("%d", result);
   return 0;
}
```

O/P

6 3 4 9

23

ii. j = (i++) + (++i)

```
#include<stdio.h>
int main()
{
   int i, j, x, y;
```

```
scanf("%d", &i);
x = i++;
y = ++i;
j = x + y;
printf("i = %d\n", i);
printf("j = %d", j);
return 0;
}
```

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i = 7

j = 12

b. Square root of a given number.

```
#include<stdio.h>
#include<math.h>
int main()
{
   float num, result;
   scanf("%f", &num);
   result = sqrt(num);
   printf("%.2f", result);
   return 0;
}
```

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5.00

(c) Find the area of circle, square, rectangle, and triangle.

Area of Circle

Formula

$$\boxed{\text{Area} = \pi r^2}$$

Where:

- **r = radius**
- **π (pi) ≈ 3.14**

//Area of Circle

```
#include<stdio.h>
#define PI 3.14
int main()
{
   float radius;
   scanf("%f", &radius);
   float area = PI * radius * radius;
   printf("Area of Circle = %.2f\n", area);
   return 0;
```

```
}
```

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Area of Circle = 78.50

Area of Square – Formula

$$\boxed{\textbf{Area} = \textbf{side}^2}$$

Where:

- **side = length of one side of the square**

```
// Area of Square
#include<stdio.h>
int main()
{
float side;
scanf("%f", &side);
float area = side * side;
printf("Area of Square = %.2f\n", area);
return 0;
}
```

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Area of Square = 16.00

Area of Rectangle – Formula

$$\boxed{\textbf{Area} = \textbf{Length} \times \textbf{Breadth}}$$

Where:

- **Length (L) = longer side**
- **Breadth (B) = shorter side**

// Area of Rectangle

```
#include<stdio.h>
int main()
{
   float length, breadth;
   scanf("%f %f", &length, &breadth);
   float area = length * breadth;
   printf("Area of Rectangle = %.2f\n", area);
  return 0;
}
```

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Area of Rectangle = 15.00

Area of Triangle – Formula

$$\boxed{\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height}}$$

Where:

- **Base (b)** = base of the triangle
- **Height (h)** = perpendicular height

//Area of Triangle

```
#include<stdio.h>
int main()
{
   float base, height, area;
   scanf("%f %f", &base, &height);
   area = 0.5 * base * height;
   printf("Area of Triangle = %.2f cm^2", area);
   return 0;
}
```

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10 6

Area of Triangle = 30.00 cm^2

d. Find the maximum of three numbers using conditional operator.

```
//Using ternary operator or conditional operator
//Biggest of Three numbers
#include<stdio.h>
int main()
{
   int a, b, c, total;
   scanf("%d %d %d", &a, &b, &c);
   total = (a>b) ? ((a>c) ? a : c) : ((b>c) ? b : c);
   printf("The Biggest Value=%d", total);
   return 0;
}
```

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30

90

50

The Biggest Value=90

e. Take marks of 5 subjects in integers, find the total in integer and average in float.

```
#include <stdio.h>
int main()
{
    int English, Maths, Science, Psychology, History;
    int total;
    float average;
    scanf("%d %d %d %d %d", &English, &Maths, &Science,
&Psychology, &History);
total = English + Maths + Science + Psychology + History;
average = total / 5.0;
printf("%d\n", total);
printf("%.2f", average);
return 0;
}
```

O/P

```
80 75 90 85 70
400
80.00
```